



Final Presentation

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Original Goals

- **Develop synthetic data generate pipeline to create “Visual Thoughts” (VCoT)**
 - Intermediate reasoning steps include visual information (coordinates, structural hierarchies, values extracted from image)
- **Improve DatBench/Perceptron 10 performance by including VCoT data in Datology Curated training mix**
 - Should see improvements in targeted domains without regressions in others
- **Why should we pursue this?**
 - Reasoning VLMs has been shown to be very powerful, we should try to induce reasoning in our models
 - MM-GEN, our synthetic data pipeline, is currently only used to create short QA samples
 - Would help with post-training greatly



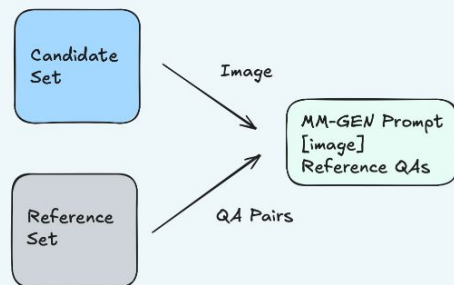
Background/Context

- **Mammoth: Large-scale multimodal instruction-tuning dataset**
 - Includes relatively large amount of reasoning data
- **Datology Curated: Mammoth + synthetic data targeting OCR, chart, and pointing tasks**
 - 75% Mammoth
 - 10% Synthetic OCR
 - 10% Synthetic Chart
 - 5% Synthetic Pointing/Counting
- **Training Run Configs:**
 - Sequence length: 4096
 - Cross document attention: Off
 - Native Resolution: On/Off depending on runs

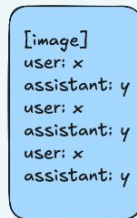


Data Generation

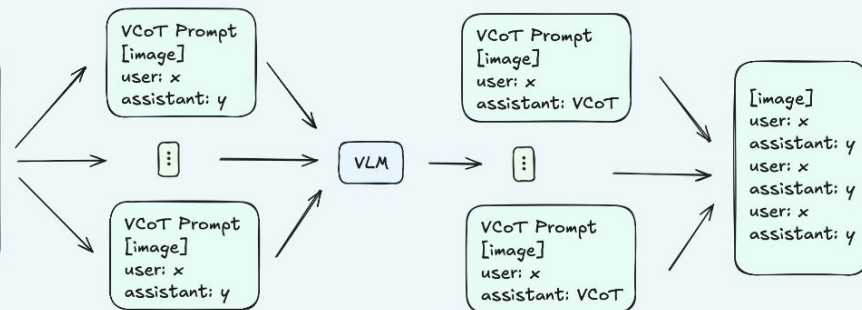
MM-GEN



Strong VLM



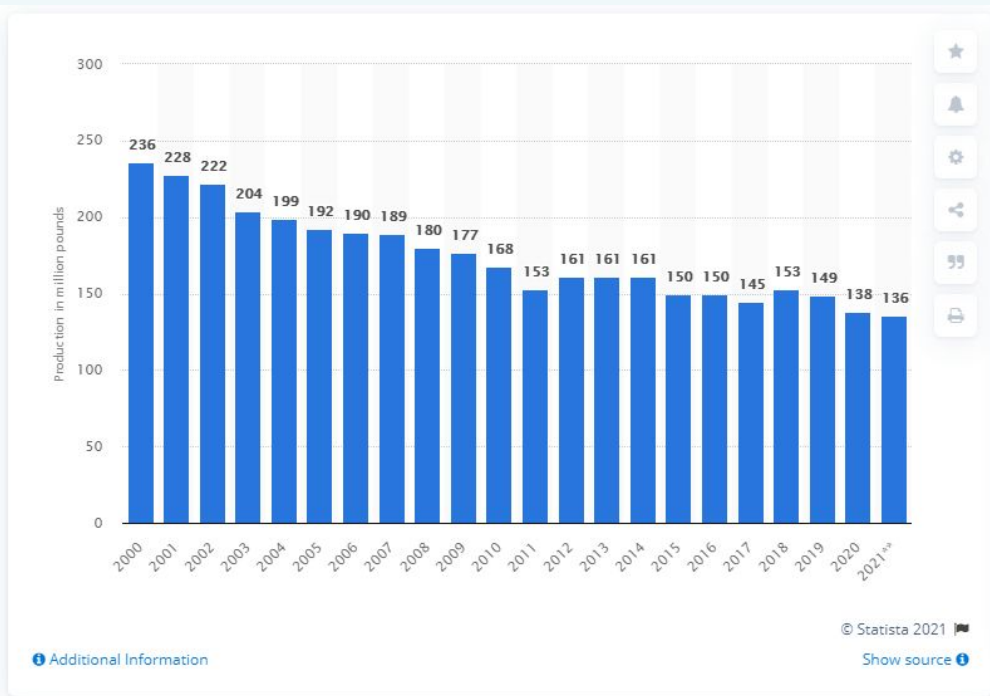
VCoT



- **VCoT 8B: 2 million multi-turn samples generated with Qwen3-VL-8B-Instruct (20m total)**
- **VCoT 32B: 1.4 million multi-turn samples generated with Qwen3-VL-32B-Instruct (7m total)**



Training Sample Refusal



Original MM-GEN QA (Incorrect):

User: What's the title of the chart?

Assistant: Reported cases of guinea worm disease, 1997

VCot: The provided chart does not have a title visible in the image. The chart displays "Production in million pounds" on the y-axis and years from 2000 to 2021 on the x-axis, with bars showing production values for each year. The source is cited as "Statista 2021" at the bottom right.

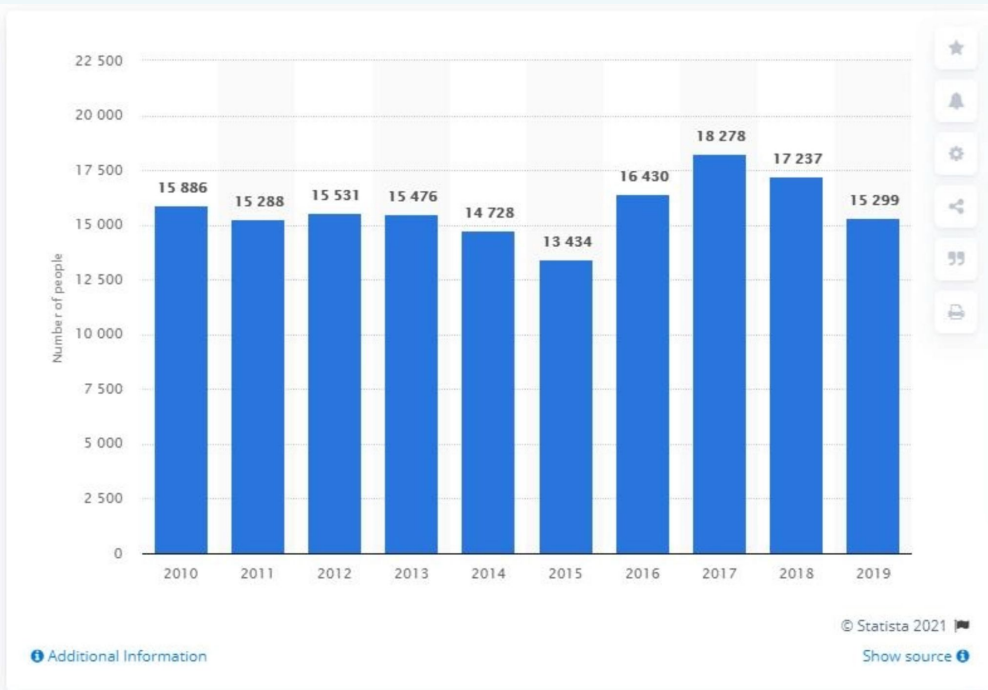
The answer provided, "Reported cases of guinea worm disease, 1997," is not supported by the chart. The chart shows production data (likely of a commodity, not disease cases), and the years range from 2000 to 2021, not 1997. There is no mention of guinea worm disease or any disease-related data.

Therefore, the chart does not support the given answer.

Answer: \boxed{\text{The chart does not have a title, and the given answer is not supported by the chart.}}



Eval Sample Refusal



Question: What was the highest number of cannabis-related crimes reported in Italy between 2010 and 2019?

Ground Truth: 18278

Output: The provided chart does not contain any information about cannabis-related crimes in Italy. It displays the "Number of people" for a specific category (likely a demographic or health metric) from 2010 to 2019.

The values shown are:

2010: 15,886
2011: 15,288
2012: 15,531
2013: 15,476
2014: 14,728
2015: 13,434
2016: 16,430
2017: 18,278
2018: 17,237
2019: 15,299

The highest value in this chart is 18,278, which occurred in 2017. However, this data is not related to cannabis-related crimes in Italy.

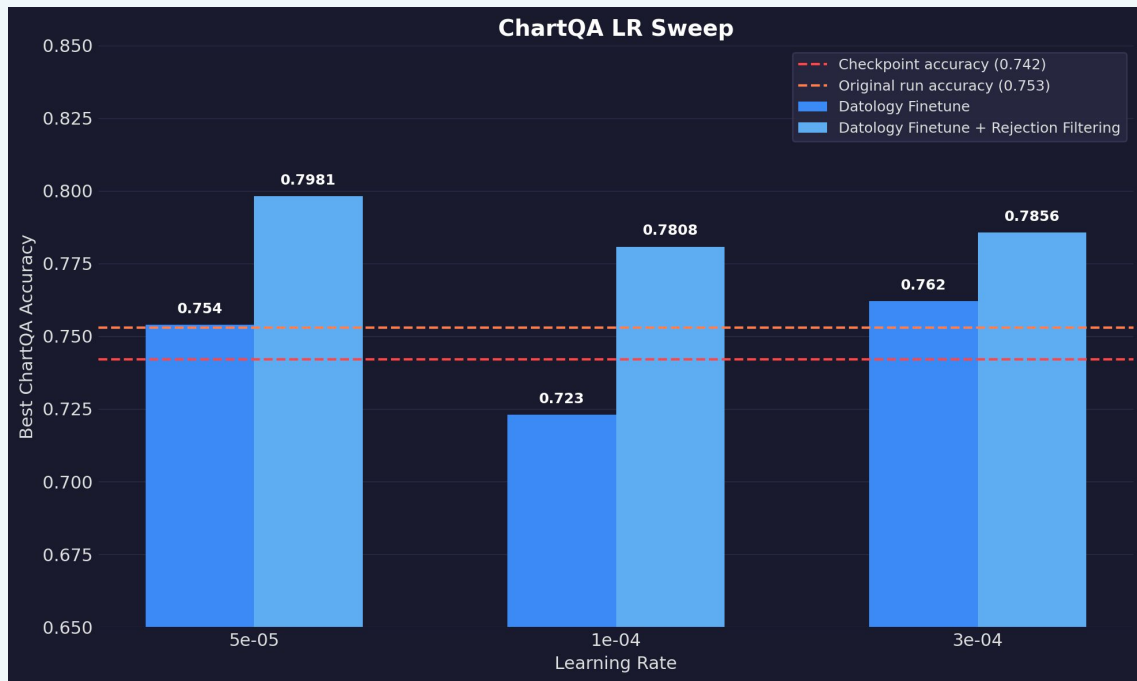
The answer provided, "18,278," is the maximum value shown in the chart, but it does not correspond to the question about cannabis-related crimes in Italy. The chart lacks any context or labels indicating it is about crimes or Italy.

Therefore, the chart does not support the given answer. The correct response is that the chart does not contain the information needed to answer the question. Answer: \boxed{The chart does not contain information about cannabis-related crimes in Italy.}



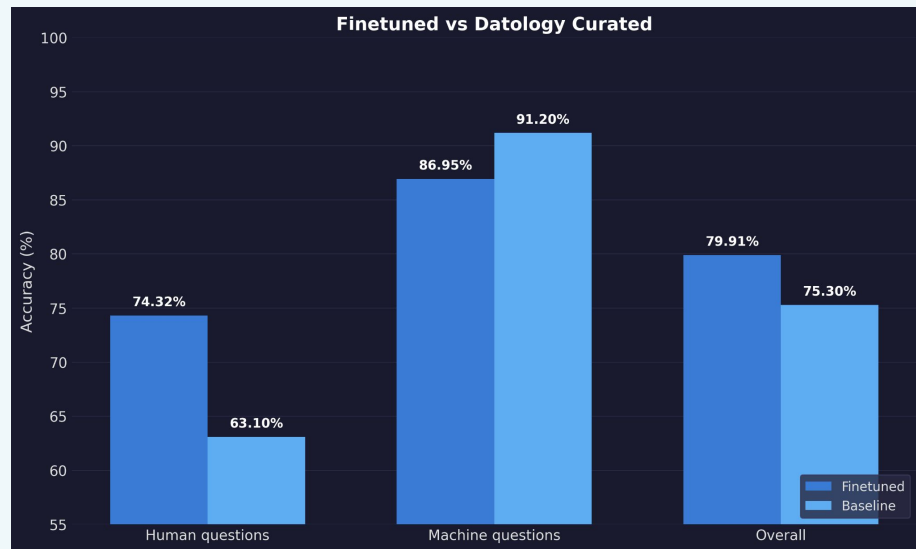
Post-Facto Reasoning Generation -> Refusals

- Low quality samples propagate through data generation
- 10% of pre-filtered eval outputs contain refusal phrases
- 3% of training data contained refusals (original QA pair incorrect, model refused to provide reasoning trace)

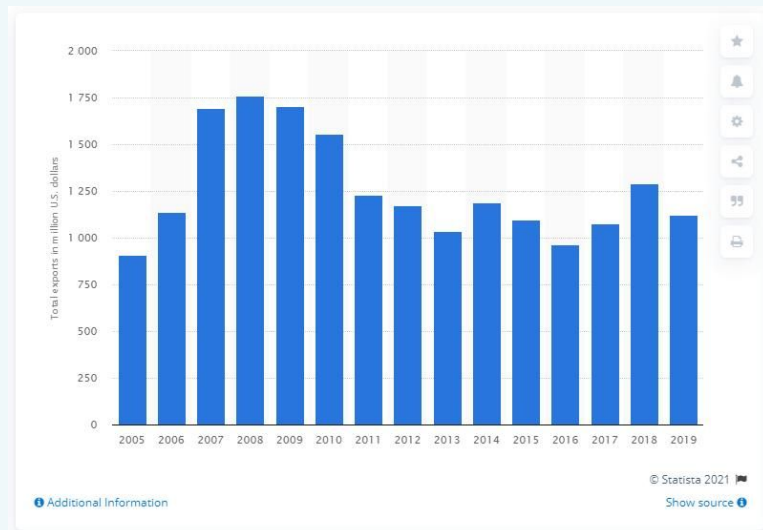


Trading Perception for Reasoning

- Reasoning helps with complex questions that require multiple steps to solve
- Model gets worse at simple perception / text extraction questions
 - Overthinks/misinterprets what the question is asking
 - Extracts text incorrectly
- Perception in OOD tasks remain untouched (OCR evals)



Where Reasoning Hurts



Question: What was the total value of merchandise exported to Canada from the U.S. in 2019?

GT: 1125

Datology: 1120.4

Finetuned::

The chart is a bar graph showing the "Total exports in million U.S. dollars" from the U.S. to Canada for each year from 2005 to 2019.

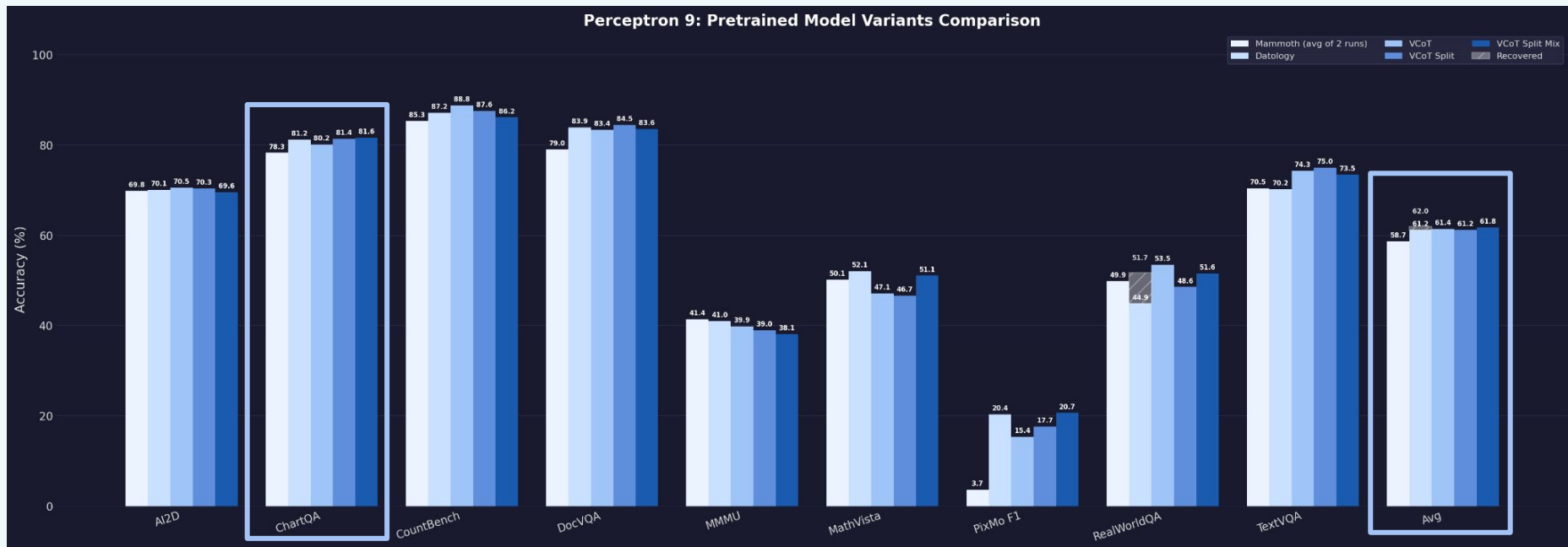
1. ****Identify the relevant part of the chart:**** To answer the question about 2019, I need to locate the bar corresponding to the year 2019 on the horizontal axis (x-axis).
2. ****Read the specific value:**** The bar for 2019 is the rightmost bar on the chart. By looking at its height relative to the vertical axis (y-axis), which is labeled "Total exports in million U.S. dollars", I can determine its value. **The bar for 2019 reaches a point that is clearly above the 1000 mark and below the 1500 mark. It appears to be approximately halfway between 1000 and 1500, which would be 1250.**
3. ****Show the reasoning:**** The y-axis has major gridlines at 0, 250, 500, 750, 1000, 1250, 1500, 1750, and 2000. **The top of the 2019 bar is positioned just below the 1500 line, but significantly above the 1250 line. A reasonable visual estimation places it at approximately 1250 million U.S. dollars.**
4. ****Compare with other years (if needed):**** While not necessary for the specific question, the bar for 2019 is visibly lower than the bars for 2008, 2009, and 2010, which are all above 1500 million U.S. dollars, and higher than the bars for 2016 and 2017, which are around 1000 million U.S. dollars. This confirms that 2019 is not the peak year, but the value is consistent with the 1250 million U.S. dollars figure.

Based on the visual evidence from the chart, the total value of merchandise exported to Canada from the U.S. in 2019 was approximately 1250 million U.S. dollars.

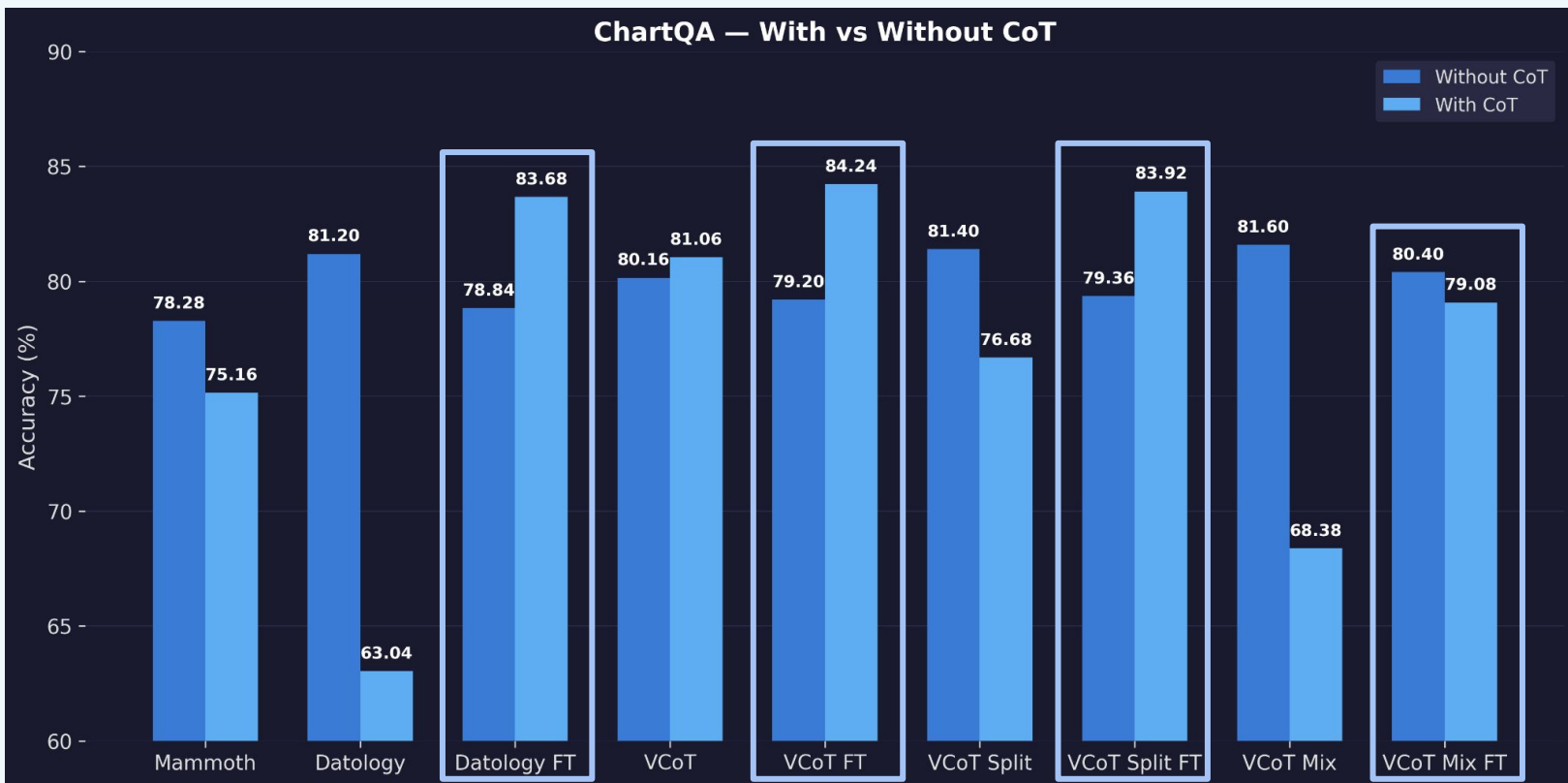
Answer: \boxed{1250}



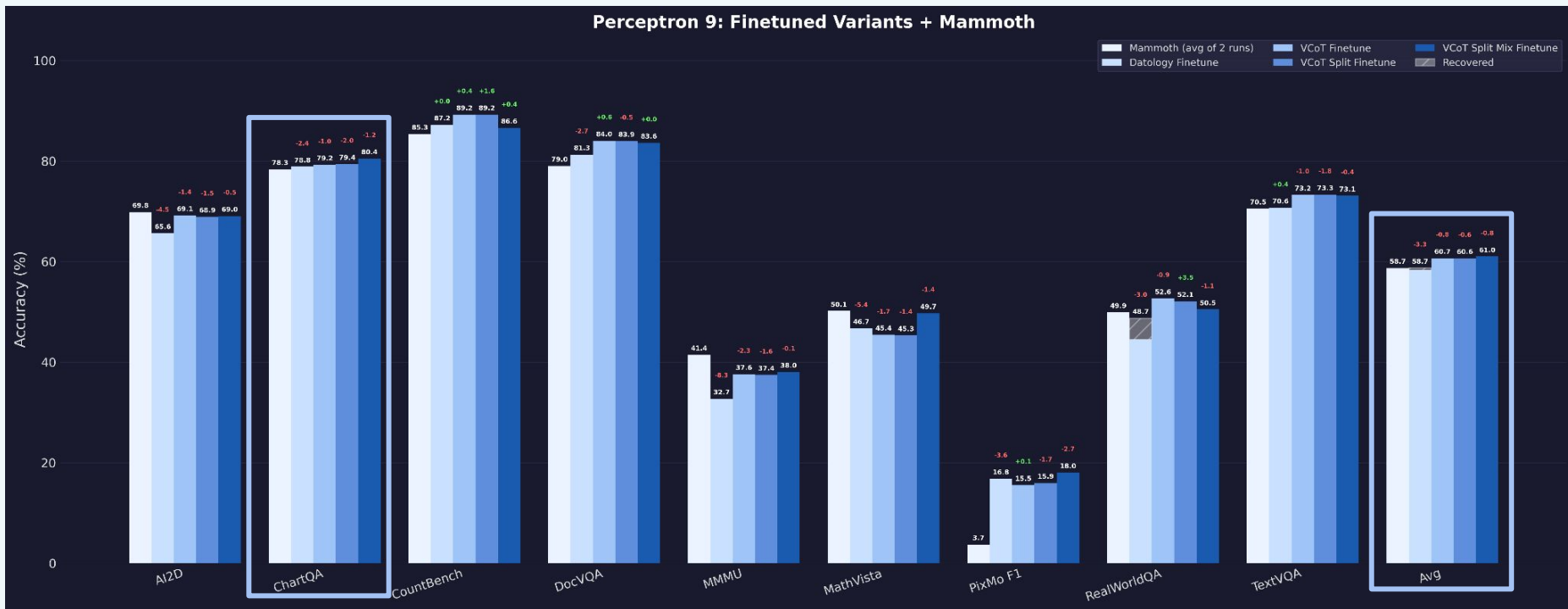
Adding Reasoning Data in Pretraining



Forced Thinking

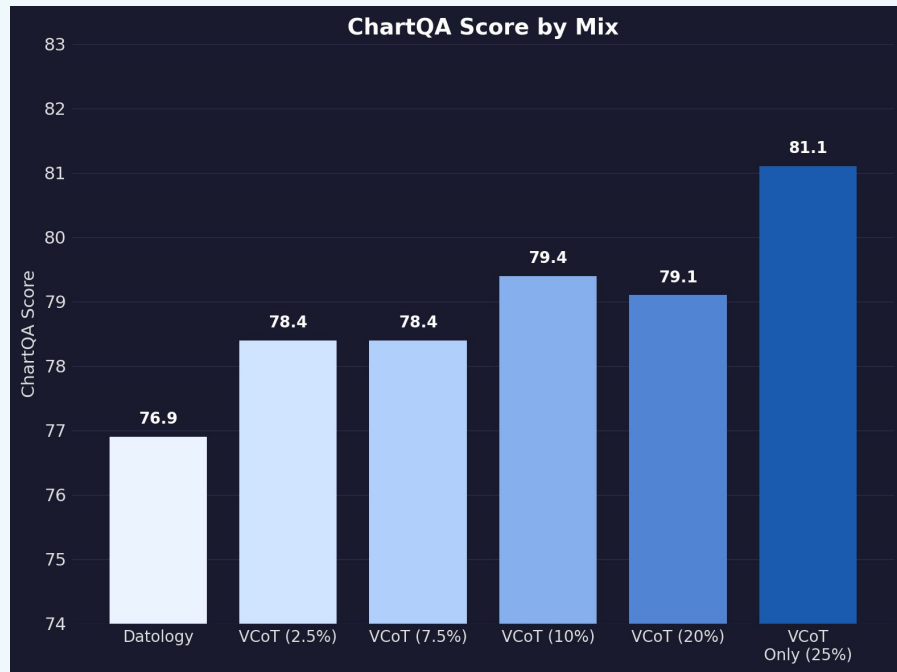


Regressions after Finetuning



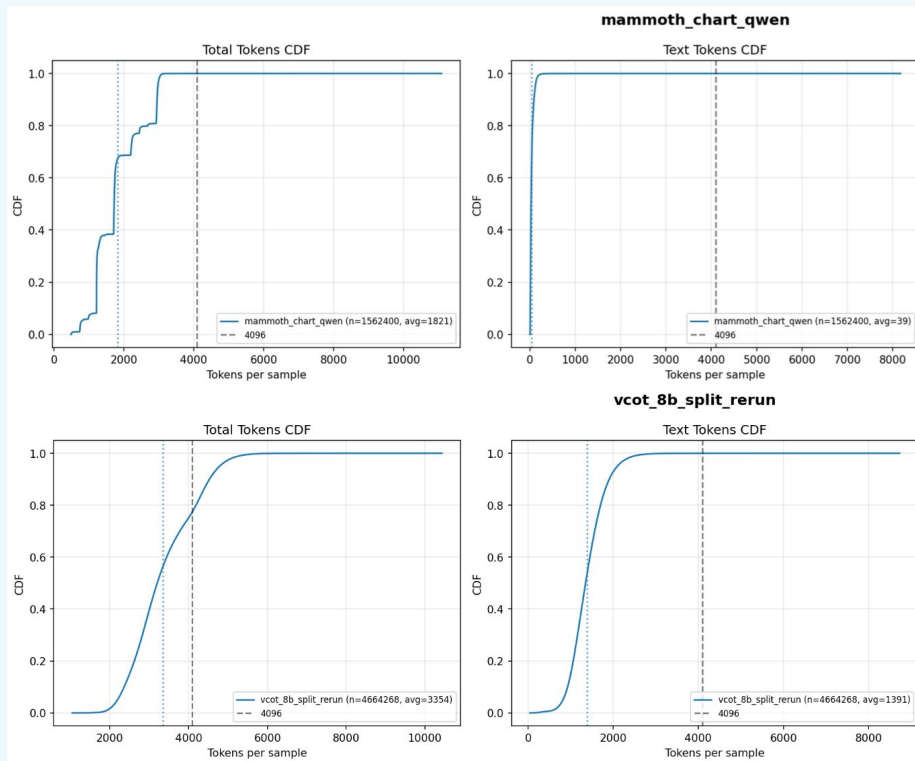
Poor Data Synergy

- Gains generally correlate with increasing the proportion of VCoT data
- Largest improvement comes from removing short form MM-GEN data
- Even only VCoT + Mammoth produces structured reasoning ~60% of the time
- Suggests VCoT data doesn't play nicely with Mammoth or MM-GEN



Poor Synergy with MM-GEN

- Reasoning traces contribute *more* to gradient updates in their batches
 - Trained models are still biased towards outputting short responses
- Effective gradient could be stronger for MM-GEN samples
 - Short QA samples tend to be more stylistically consistent and are lower entropy → lower gradient variance → easier to learn
 - Longer reasoning traces are higher entropy and have semantic nuance → higher gradient variance → harder to learn



Poor Synergy with Mammoth

- **Team had previously trained a model that reasoned all of the time**
 - Rephrased Mammoth to use the reasoning format + lacked short form MM-GEN data
- **Mammoth already includes reasoning data, wide variety in format/structure**
 - Creates noisy gradients
- **Majority of training data might need to match in structure for model to easily default to reasoning**

Rephrased Mammoth:

EXTRACT: The train's movement may cause water vibrations, cast shadows/noise, and pose collision risks to the boat.

REASON: These factors directly impact boat stability, navigation systems, and safety during bridge crossings.

MATH: None

ANSWER: Stability, visibility, and safety risks

Eval output:

EXTRACT: The average cotton production of the top three countries is 5336.67 thousand metric tons.

REASON: The average is calculated by summing the top three values and dividing by three.

MATH: $(5933 + 4316 + 3316) / 3 = 5336.67$

ANSWER: 5336.67



Problems with Mixing Domain Specific Data

- **Model trained on mix of MM-GEN + VCoT chart data never outputs reasoning**
 - Even finetuned variant does not output CoT, even when prompted to
- **Loss is lower when including VCoT, suggesting model *is* learning reasoning**
 - Bimodal output distribution doesn't condition model to output reasoning
- **Continuation prompting (ending eval prompts with common VCoT starting phrases) induces reasoning**

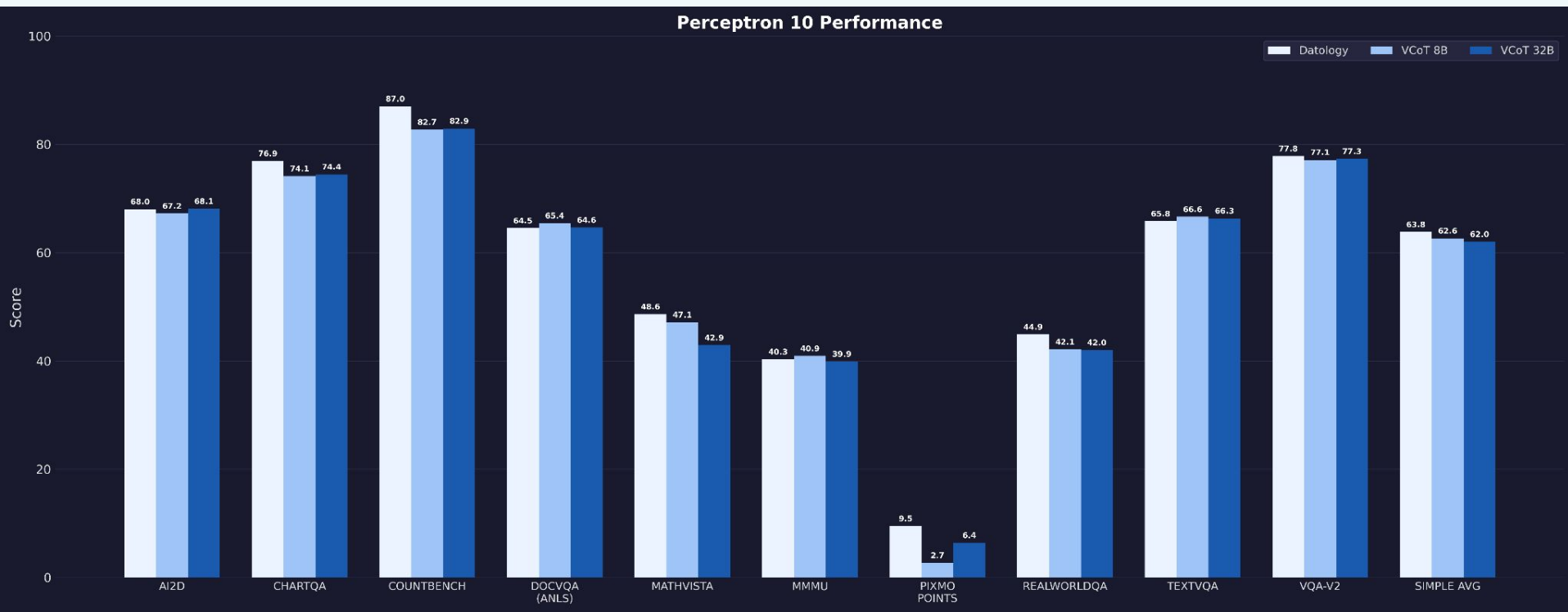


How to Learn to Reason

- **10 to 20% structured reasoning data isn't enough for models to default to reasoning**
 - Specialized instructions can help unlock latent reasoning abilities, but not standard
- **Unlocking reasoning in pretraining**
 - Rephrase Mammoth to use desired reasoning structure
 - Train on rephrased mammoth, synthetic reasoning, and short MM-GEN samples
 - Model would default to reasoning, but should learn to output short answers for MM-GEN tasks (due to gradients being clearer)
 - Could use <think> and <no_think> tags
- **Unlocking reasoning in post-training**
 - Create reasoning traces for wide variety of tasks
 - Include reasoning data in pretraining, use RL to induce reasoning more frequently



Additional Insights: Does Model Size Matter?



Additional Insights: Finetuning as a Proxy?

